

A Real Parseval Smooth-Connected Subcase of Full-Spark Genericity

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Status. This packet records a likely valid partial result for the closing question in Needham–Shonkwiler, *Toric Symplectic Geometry and Full Spark Frames*, arXiv:2110.11295. The result is new work in this run, but it is deliberately scoped: it handles smooth connected real Parseval prescribed-norm frame spaces, not arbitrary real fixed-spectrum spaces and not quaternionic frame spaces.

1 Source Question

Needham–Shonkwiler prove that, for complex frames with prescribed column norms r and prescribed frame spectrum λ , full-spark frames have full measure exactly in the strongly admissible case. In Section 5, after the algebraic proof sketch, they write:

such an algorithm would provide endless quantities of full spark FUNTFs.

It is natural to ask whether the analog of [Theorem 1.1](#) holds for real and for quaternionic frames. Symplectic geometry is not the right tool in either case, but it is very plausible that the algebraic proof sketched in [Section 4.2](#) could be adapted to the real case to show that full spark frames are dense in real frame spaces. In a different direction, the perspective based on isotropy orbits and isoparametric submanifolds introduced in [\[44\]](#) seems like the most promising way to understand the measures on real and quaternionic frame spaces.

Finally, the spaces $\mathcal{F}_{\lambda}^{d,N}(\mathbf{r})/(\mathrm{U}(d) \times G)$ are examples of *weight varieties* [\[35\]](#), and Goldin [\[22\]](#) has determined the rational cohomology ring of certain weight varieties, including the quotient $\mathcal{F}_{(\lambda,\dots,\lambda)}^{d,N}(\mathbf{r})/(\mathrm{U}(d) \times G)$ of the space of λ -tight frames with fixed frame vector norms whenever it is a manifold. In particular, this determines the rational cohomology ring of the $(\mathrm{U}(d) \times G)$ -quotient of FUNTF space when N and d are relatively prime. What about in the non-manifold case or for more general frame spectra?

Thus the extracted target is the real/quaternionic analogue of their complex full-spark genericity theorem. The packet below gives a real Parseval subcase.

2 Definitions

For $r = (r_1, \dots, r_N) \in (0, 1)^N$ with $\sum_i r_i = d$, set

$$\mathrm{PF}_d^{\mathbb{R}}(r) = \{F = [f_1 | \dots | f_N] \in \mathbb{R}^{d \times N} : FF^T = I_d, \|f_i\|^2 = r_i \text{ for all } i\}.$$

This is the real Parseval prescribed-norm subcase of the frame spaces in arXiv:2110.11295, corresponding to frame operator I_d and spectrum $(1, \dots, 1)$. A frame F is *full spark* if every $d \times d$ column minor is nonzero. Equivalently,

$$\Delta(F) := \prod_{\substack{I \subset \{1, \dots, N\} \\ |I|=d}} \det(F_I)$$

is nonzero.

3 Proof Intuition

The real obstruction is not existence: for any interior Parseval diagonal r , one can start with any full-spark real d -plane and scale its coordinates by a positive diagonal matrix until the orthogonal projection onto the scaled plane has diagonal r . The log-sum-exp form of the torus moment map sends the positive diagonal orbit of a full-spark plane onto the interior of the hypersimplex.

Once one full-spark point exists, the only remaining issue is whether the determinant product Δ could vanish on a whole component of the real frame space. On a connected smooth real-analytic frame space, this cannot happen: a nonzero real-analytic function has a zero set of measure zero. Hence connected smooth real Parseval spaces inherit full-spark density. Mare's real Schur–Horn connectedness theorems provide concrete nonconstant families where the connectedness hypothesis is known.

4 The Partial Theorem

Theorem 1 (smooth connected real Parseval subcase). *Let $2 \leq d < N$, and let $r \in (0, 1)^N$ satisfy $\sum_i r_i = d$. Suppose that $\text{PF}_d^{\mathbb{R}}(r)$ is a connected smooth real-analytic submanifold of $\mathbb{R}^{dN}$. Then the full-spark locus*

$$\{F \in \text{PF}_d^{\mathbb{R}}(r) : \Delta(F) \neq 0\}$$

is nonempty, open, dense, and has full Hausdorff measure in $\text{PF}_d^{\mathbb{R}}(r)$.

Proof. We first show nonemptiness. Choose a full-spark real d -plane $L_0 \subset \mathbb{R}^N$; for instance, take the row span of a $d \times N$ Vandermonde matrix with distinct real nodes. Let $p_I(L_0)$ be its Pluecker coordinates, all nonzero.

For $u = (u_1, \dots, u_N) \in \mathbb{R}^N$, let $D_u = \text{diag}(e^{u_1}, \dots, e^{u_N})$ and $L_u = L_0 D_u$. The diagonal of the orthogonal projection onto L_u is the Grassmannian moment map

$$\mu(u) = \frac{\sum_{|I|=d} |p_I(L_0)|^2 e^{2 \sum_{i \in I} u_i} e_I}{\sum_{|I|=d} |p_I(L_0)|^2 e^{2 \sum_{i \in I} u_i}},$$

where $e_I = \sum_{i \in I} e_i$. This map is the gradient of

$$\Phi(u) = \log \left(\sum_{|I|=d} |p_I(L_0)|^2 e^{2 \langle u, e_I \rangle} \right),$$

up to the harmless factor $1/2$. Since the support of the exponential sum is the full vertex set of the hypersimplex

$$\Delta(d, N) = \text{conv}\{e_I : |I| = d\},$$

the gradient map sends $\mathbb{R}^N / \mathbb{R}(1, \dots, 1)$ onto the relative interior of $\Delta(d, N)$. Indeed, after restricting to the hyperplane $\sum_i u_i = 0$, Φ is strictly convex modulo constants, and as u diverges in any direction its gradient approaches the exposed face maximizing that direction; hence the image is precisely the interior of the convex hull of the support.

The assumption $0 < r_i < 1$ and $\sum_i r_i = d$ says exactly that r lies in the relative interior of $\Delta(d, N)$. Therefore there is u with $\mu(u) = r$. Positive diagonal scaling preserves nonvanishing of

all Pluecker coordinates, so L_u is still full spark. Taking any orthonormal row matrix F spanning L_u , we get $F^T F = P_{L_u}$, hence $FF^T = I_d$, $\|f_i\|^2 = (P_{L_u})_{ii} = r_i$, and F is full spark. Thus Δ is not identically zero on $\text{PF}_d^{\mathbb{R}}(r)$.

The function Δ is a polynomial on $\mathbb{R}^{dN}$, hence its restriction to the assumed connected smooth real-analytic manifold $\text{PF}_d^{\mathbb{R}}(r)$ is real analytic. Since it is not identically zero, its zero set has empty interior and Hausdorff measure zero in that manifold by the standard zero-set theorem for nontrivial real-analytic functions. The complement is therefore nonempty, open, dense, and full measure. $\square$

5 Concrete Nonconstant Families

The theorem becomes a verified partial answer whenever known results give connectedness and smoothness. Mare studies the Schur–Horn map

$$\text{Gr}_d(\mathbb{R}^N) \rightarrow \mathbb{R}^N, \quad L \mapsto \text{diag}(P_L),$$

and proves connectivity criteria for its real fibers and for the corresponding prescribed-norm real Parseval frame spaces. The standard fixed-operator frame-variety criterion says that $\text{PF}_d^{\mathbb{R}}(r)$ is smooth whenever no nonempty proper subset $J \subset \{1, \dots, N\}$ has $\sum_{j \in J} r_j \in \{1, \dots, d-1\}$; such integer subset sums are the trace obstruction forced by an orthodecomposable singularity.

Corollary 2 (a paired nonconstant family). *Let $N = 2p$, $d = 2q$, and $1 \leq q$ with $2q < p$. Choose*

$$\frac{q}{2p-2q} < \beta < \frac{q}{p}, \quad \alpha = 1 - \frac{p-q}{q}\beta,$$

and set

$$r = (\underbrace{\alpha, \dots, \alpha}_q, \underbrace{\beta, \dots, \beta}_{p-q}, \underbrace{\alpha, \dots, \alpha}_q, \underbrace{\beta, \dots, \beta}_{p-q}).$$

If β is chosen outside the finite exceptional set for which a proper subset sum of the entries of r is an integer, then full-spark frames have full Hausdorff measure in $\text{PF}_d^{\mathbb{R}}(r)$.

Proof. Mare’s paired even- N criterion in Section 4 applies to this vector: the interval for β is exactly the nonconstant interior of the range in which the corresponding real prescribed-norm Parseval frame space is connected. The displayed finite-exception condition gives smoothness by the fixed-operator frame-variety criterion. The entries of r lie in $(0, 1)$ and sum to $2q = d$. The conclusion follows from Theorem 1. $\square$

6 Verification and Novelty Notes

The run indexes were searched for arXiv id 2110.11295 and for the phrases “full spark”, “real frame spaces”, “quaternionic frames”, “Parseval frames”, and “Schur–Horn”. No existing packet for this paper or this real full-spark-density subcase was found. Nearby durable memory includes a 2108.02275 packet using Caine–Needham–Shonkwiler 2025 for real Parseval connectedness, but that packet concerns connectedness, not full-spark genericity.

External literature checked during this loop included Needham–Shonkwiler 2022, Cahill–Mixon–Strawn 2017, Mare 2024, and Caine–Needham–Shonkwiler 2025. Cahill–Mixon–Strawn already prove generic full spark for real equal-norm tight frames via irreducibility. Mare and Caine–Needham–Shonkwiler provide real Parseval connectedness tools, but neither was found to state

the full-spark-density conclusion for nonconstant prescribed norms. The novelty confidence for the abstract criterion is moderate: it is elementary once the smooth connected hypotheses are isolated, so it may be folklore, but it was not found as a recorded answer to the 2110.11295 closing question.

7 Limitations

This does not solve the full real analogue of Needham–Shonkwiler Theorem 1.1. It does not treat non-Parseval prescribed spectra, real spaces that are singular or disconnected, or the measure theory of quaternionic frame spaces. The connectedness and smoothness assumptions are genuine: without them, a determinant polynomial could in principle vanish on an entire component or stratum, so the argument would not prove full-measure genericity.

8 Human Review Recommendation

Review the log-sum-exp moment-map surjectivity step and the invocation of the real-analytic zero-set theorem. Then check the cited smoothness criterion for the exact normalization $FF^T = I_d$. If those pass, the packet should be accepted as a partial real Parseval subcase and not as a full solution of the source question.

References

- [1] T. Needham and C. Shonkwiler, *Toric Symplectic Geometry and Full Spark Frames*, Applied and Computational Harmonic Analysis 58 (2022), 409–445; arXiv:2110.11295.
- [2] J. Cahill, D. G. Mixon, and N. Strawn, *Connectivity and irreducibility of algebraic varieties of finite unit norm tight frames*, SIAM Journal on Applied Algebra and Geometry 1 (2017), 38–72.
- [3] A.-L. Mare, *Connectivity properties of the Schur-Horn map for real Grassmannians*, Abhandlungen aus dem Mathematischen Seminar der Universitaet Hamburg 94 (2024), 33–55; arXiv:2309.11596.
- [4] N. Strawn, *Finite frame varieties: nonsingular points, tangent spaces, and explicit local parameterizations*, Journal of Fourier Analysis and Applications 17 (2011), 821–853.
- [5] A. Caine, T. Needham, and C. Shonkwiler, *Optimization and the Topology of Spaces of Parseval Frames*, arXiv:2505.14860, 2025.